

Quiz 1 — Practice Problems

Reference: Munson, B. R., Rothmayer, A. P., Okiishi, T. H., & Huebsch, W. W. (2013). *Fundamentals of fluid mechanics* (7th ed.). John Wiley & Sons.

How to work through it

- Work in SI throughout, as in the lectures. Where the textbook poses a problem in British Gravitational units, convert the data first and answer in SI.
- Take fluid properties (ρ , μ , ν , σ , E_v , p_v) from the tables in the textbook, or from the tables reproduced in the lectures.
- Attempt each problem yourself first before looking anything up. Getting stuck, and then unsticking yourself, is the way to learn.
- Estimate the order of magnitude of the answer before you compute it, and check it afterwards. This is an important step in understanding concepts and parameter spaces.
- Submission is not needed. This is for you to practice for quiz 1.

Part A — Conceptual Questions

Read the *Conceptual Questions* on page 31 of the textbook and answer **1.1C**, **1.2C**, **1.3C**, **1.4C** and **1.5C**. Choose the correct answer and write 1-2 sentences justifying the choice, and say what makes each of the other options wrong.

Part B — Practice Problems

Solve the following problems from the textbook:

1. **Problem 1.46** — the Martian atmosphere.
2. **Problem 1.51** — oxygen in a tank.
3. **Problem 1.58** — capillary-tube viscometer.
4. **Problem 1.67** — Reynolds numbers.
5. **Problem 1.74** — viscosity.
6. **Problem 1.76** — Newtonian or non-Newtonian fluid.
7. **Problem 1.78** — Shaft pulling.
8. **Problem 1.97** — compressibility of fluids.
9. **Problem 1.105** — isentropic bulk modulus.
10. **Problem 1.115** — cavitation?
11. **Problem 1.124** — surface tension.

If you want to solve more problems of one topic: surface tension, 1.118 to 1.123 and 1.126 to 1.130; cavitation and vapor pressure, 1.111 to 1.116; viscosity in rotating geometries, 1.85 to 1.89; compressibility, 1.100 to 1.107.